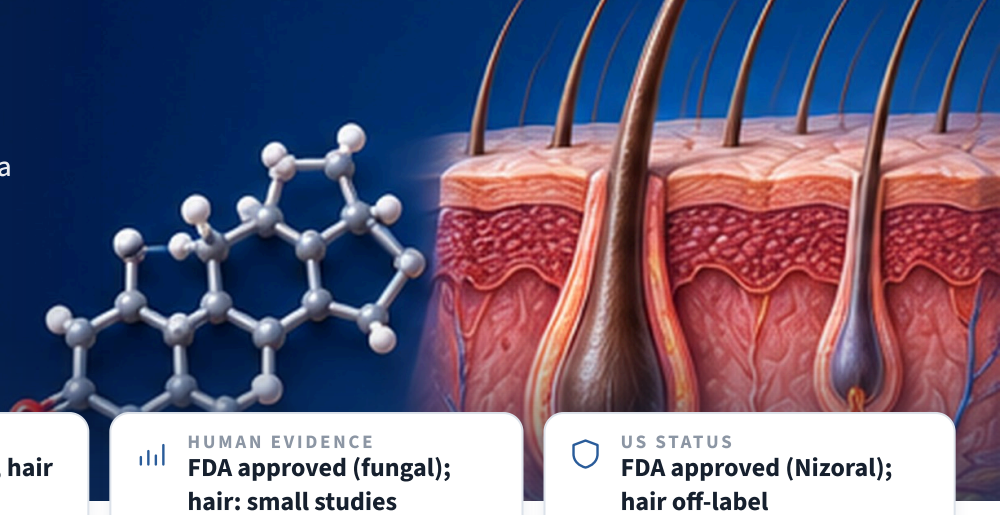
HAIR LOSS
04

Ketoconazole

Azole antifungal used on the skin and scalp as a shampoo or cream



CLASS
Antifungal

STUDIED FOR
Dandruff, fungal skin, hair loss

HUMAN EVIDENCE
FDA approved (fungal); hair: small studies

US STATUS
FDA approved (Nizoral); hair off-label

STUDIED FOR

Dandruff, seb. derm.	Fungal skin infection	Male pattern loss
HUMAN	HUMAN	HUMAN + ANIMAL

Hair use is off-label. A 2020 review found five small human studies and two mouse studies, and called for randomized controlled trials.

HOW IT WORKS IN THE BODY

Ketoconazole → Ergosterol → Fungal membrane → Less Malassezia

- **Impairs ergosterol synthesis,** a fungal membrane part, in lab studies
- **Seborrheic dermatitis:** label links effect to fewer M. ovale, unconfirmed
- **Mouse studies** found more hair regrowth than in untreated mice
- **For hair loss in people,** the mechanism is not established

EVIDENCE & WHAT'S KNOWN

EVIDENCE

FDA APPROVED
LEVEL 5 OF 5

- Approved shampoos and cream for fungal skin
- Hair: one controlled study (1998), small
- 2020 review: 5 human studies, 318 people

NOT YET ESTABLISHED

- Hair effect in randomized trials
- Long-term scalp use for hair
- Compounded mixed sprays

QUICK PRACTICAL INFO

HALF-LIFE
Not established for scalp use; not found in plasma

ROUTES STUDIED
On the skin and scalp (shampoo, cream); tablets restricted

PHARMACOKINETICS
Not detected in plasma after months of shampooing

SOLD AS
OTC 1% shampoo (Nizoral); Rx 2% shampoo and cream

MONITORED IN TRIALS
Hair density, hair diameter, scalp oil (studies)

SPORT (WADA)
Not on the 2026 list

REGULATORY

FDA	Approved for fungal skin and dandruff; not hair
FDA	Tablets: boxed warning (liver, heart rhythm)
EMA	Oral antifungal use suspended 2013; topical kept
WADA	Not on the 2026 list

REFERENCES

1. FDA, ketoconazole shampoo 2% label (DailyMed), Jul 2026
2. Fields et al., Dermatol Ther, topical ketoconazole for hair loss (review), Jan 2020
3. EMA, ketoconazole-containing medicines referral, Jun 2014

Adonis Men's Health · checked Nov 2026. Research and regulator records only. Not medical advice.

adonismenshealth.com